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CFG Lab 6

Task 1:

$S \rightarrow A\ B$
$A \rightarrow a\ A \mid \varepsilon$
$B \rightarrow b\ B \mid c$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 5
Enter productions:
S=AB
A=aA
A=$
B=bB
B=c

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c }
FOLLOW(S) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b c }
FOLLOW(B) = { $ }
Continue? (y/n): n

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Process exited after 16.25 seconds with return value 0
Press any key to continue . . .
```

Task 2:

$E \rightarrow T\ R$
$R \rightarrow +\ T\ R \mid \varepsilon$

$T \rightarrow F\ Y$

$Y \rightarrow *\ F\ Y \mid \varepsilon$

$F \rightarrow (\ E\ ) \mid id$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 8
Enter productions (Example: E=TR):
E=TR
R=+TR
R=$
T=FY
Y=*FY
Y=$
F=(E)
F=id

Find FIRST and FOLLOW of: F
FIRST(F) = { ( i }
FOLLOW(F) = { * + $ ) }
Continue? (y/n): y

Find FIRST and FOLLOW of: Y
FIRST(Y) = { * $ }
FOLLOW(Y) = { + $ ) }
Continue? (y/n): N

--------------------------------
Process exited after 120.9 seconds with return value 0
Press any key to continue . . .
```

Task 3:

$S \rightarrow A\ B\ C$

$A \rightarrow a \mid \varepsilon$

$B \rightarrow b \mid \varepsilon$

$C \rightarrow$ **c**

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 6
Enter productions (Example: E=TR):
S=ABC
A=a
A=$
B=b
B=$
C=c

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c }
Continue? (y/n): y

Find FIRST and FOLLOW of: S
FIRST(S) = { a b c }
FOLLOW(S) = { $ }
Continue? (y/n): n

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Process exited after 33.23 seconds with return value 0
Press any key to continue . . .
```

Task 4:

$$S \rightarrow a\ S \mid b\ A$$
$$A \rightarrow c\ A \mid d$$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 4
Enter productions (Example: E=TR):
S=aS
S=bA
A=cA
A=d

Find FIRST and FOLLOW of: A
FIRST(A) = { c d }
FOLLOW(A) = { $ }
Continue? (y/n): n

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Process exited after 44.51 seconds with return value 0
Press any key to continue . . .
```

Task 5:

$$S \rightarrow (S) S \mid \varepsilon$$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 2
Enter productions (Example: E=TR):
S=(S)S
S=$

Find FIRST and FOLLOW of: S
FIRST(S) = { ( $ }
FOLLOW(S) = { $ ) }
Continue? (y/n): n

--------------------------------
Process exited after 10.35 seconds with return value 0
Press any key to continue . . .
```

Task 6:

$$S \rightarrow A\ B$$
$$A \rightarrow a \mid \varepsilon$$
$$B \rightarrow b\ C$$
$$C \rightarrow c \mid \varepsilon$$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 6
Enter productions (Example: E=TR):
S=AB
A=a
A=$
B=bC
C=c
C=$

Find FIRST and FOLLOW of: C
FIRST(C) = { c $ }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: A
FIRST(A) = { a $ }
FOLLOW(A) = { b }
Continue? (y/n): n

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Process exited after 39.21 seconds with return value 0
Press any key to continue . . .
```

Task 7:

$E \rightarrow T\ E'$

$E' \rightarrow +\ T\ E' \mid \varepsilon$

$T \rightarrow F\ T'$

$T' \rightarrow *\ F\ T' \mid \varepsilon$

$F \rightarrow id \mid (\ E\ )$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 8
Enter productions (Example: E=TE'):
E=TE'
E'=+TE'
E'=$
T=FT'
T'=*FT'
T'=$
F=id
F=(E)

Find FIRST and FOLLOW of: F
FIRST(F) = { id ( }
FOLLOW(F) = { * + $ ) }
Continue? (y/n): n

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Process exited after 13.87 seconds with return value 0
Press any key to continue . . .
```

Task 8:

S → A B

A → B a | ε

B → b | ε

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 5
Enter productions (Example: E=TR):
S=AB
A=Ba
A=$
B=b
B=$

Find FIRST and FOLLOW of: S
FIRST(S) = { b a $ }
FOLLOW(S) = { $ }
Continue? (y/n): n

--------------------------------
Process exited after 31.59 seconds with return value 0
Press any key to continue . . .
```

Task 9:

$S \rightarrow A\ B\ C$

$A \rightarrow a\ A \mid \varepsilon$

$B \rightarrow b\ B \mid \varepsilon$

$C \rightarrow c\ C \mid d$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 7
Enter productions (Example: E=TR):
S=ABC
A=aA
A=$
B=bB
B=$
C=cC
C=d

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { $ }
Continue? (y/n): y

Find FIRST and FOLLOW of: B
FIRST(B) = { b $ }
FOLLOW(B) = { c d }
Continue? (y/n): n

--------------------------------
Process exited after 58.09 seconds with return value 0
Press any key to continue . . .
```

Task 10:

$S \rightarrow A\ C\ B$

$A \rightarrow a\ A \mid \varepsilon$

$B \rightarrow b\ B \mid \varepsilon$

$C \rightarrow c \mid d$

Output:

```
D:\Semester 6\CC Lab\cfg lab

Enter number of productions: 7
Enter productions (Example: E=TR):
S=ACB
A=aA
A=$
B=bB
B=$
C=c
C=d

Find FIRST and FOLLOW of: C
FIRST(C) = { c d }
FOLLOW(C) = { b $ }
Continue? (y/n): n

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Process exited after 11.66 seconds with return value 0
Press any key to continue . . .
```